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Instructions

This quiz will use the Coursera Lab Jupyter notebook "Lab to implement open-loop state estimation and demonstrate bad initialization" from this lesson.

In another browser tab or window, open the lab. Run the lab and enter results from the lab as your responses to the final two questions in this practice quiz.

1. What are properties of Kalman-filter state estimators versus open-loop estimators? (Select all that apply.)

1 / 1 point

- ☐ Kalman filters require fewer computational operations per measurement than open-loop estimators require.
- ☒ Kalman-filter state estimators have more confidence in their estimates (narrower confidence bounds) than open-loop state estimators.

Correct
Yes. This is correct.

- ☒ Kalman-filter state estimators are able to recover from a poor initial state estimate whereas open-loop state estimators never recover from poor initialization.

Correct
Yes. This is correct.

- ☒ Kalman-filter state-estimation errors look noisier (have less structure or correlation) than open-loop state-estimation errors.

Correct
Yes. This is correct.

2. Run the open-loop state-estimator code from the Coursera Lab. What is the final 3σ error bound on the velocity state (at $t = 100$)? Enter your answer with three digits to the right of the decimal point.

1 / 1 point

0.021016

Correct
Yes. That is correct.

3. Run the Kalman-filter state-estimator code from the Coursera Lab. During the simulation, how many times did the magnitude of the velocity error exceed the 3σ bounds on velocity error?

1 / 1 point

1

Correct
Yes. That is correct.